Arista Access Node IND-ACC-16 Hardware/Environmental Fault Recovery SOP

Node: IND-ACC-16 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_8415

Description

This Standard Operating Procedure (SOP) outlines the steps to recover from a hardware/environmental fault on Arista Access Node IND-ACC-16, which is located at the Indiranagar Exchange. The fault is identified by Alarm Code ALARM_8415 and has a severity of Warning. The failure domain is attributed to Hardware/Environmental Fault, which is characterized by ASIC memory parity errors, line card crashes, or power supply redundancy loss. On Arista Access Nodes, the role of the node is to provide access to the network, and as such, it is critical to troubleshoot and resolve this issue promptly to ensure network stability and uptime. The underlying protocol or hardware mechanics that lead to this failure domain are related to the ASIC memory parity errors, which can occur due to a variety of reasons such as hardware defects, software bugs, or environmental factors such as temperature or humidity. The line card crashes can occur due to software bugs, hardware defects, or power supply issues. The power supply redundancy loss can occur due to a variety of reasons such as hardware failure, power grid issues, or environmental factors such as temperature or humidity. To troubleshoot this issue, the network engineer must follow a structured approach that involves gathering information about the fault, isolating the root cause, and implementing a resolution. This SOP provides a step-by-step guide to troubleshooting and resolving the issue.

Pre-requisite Checks

- Verify that the Indiranagar Exchange is accessible and dispatch remote hands if necessary.
- Check the redundant link status to ensure that it is functioning properly.
- Verify that the node is powered on and all fans are spinning.
- Verify that the temperature and humidity levels are within the recommended ranges.

Potential Root Causes

Cause	Probability
ASIC Memory Parity Errors	High
Line Card Crashes	Medium
Power Supply Redundancy Loss	Low

Diagnostic Commands & Expected Outputs

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Command: show system resource, Expected Output: System Resource
Information, Description: This command provides information about the
system resources such as CPU, memory, and disk usage. It helps to
identify any resource bottlenecks that may be contributing to the
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fault.

Command: show system internal error, Expected Output: System Internal Error Information, Description: This command provides information about the internal system errors such as ASIC memory parity errors. It helps to identify the root cause of the fault.

Command: show system power supply, Expected Output: System Power Supply Information, Description: This command provides information about the power supply configuration and status. It helps to identify any power supply issues that may be contributing to the fault.
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Resolution Steps

Step 1: Step: Replace the faulty ASIC, Command: system reload no-wait, Description: This step involves replacing the faulty ASIC with a new one. The system reload command reloads the system without waiting for user confirmation.

Step 2: Step: Replace the faulty line card, Command: interface / no shutdown, Description: This step involves replacing the faulty line card with a new one. The interface command is used to configure the interface, and the no shutdown command is used to enable the interface.

Step 3: Step: Replace the faulty power supply, Command: power supply enable, Description: This step involves replacing the faulty power supply with a new one. The power supply command is used to configure the power supply, and the enable command is used to enable the power supply.

Post-Check Verification

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Command: show system resource, Expected Output: System Resource Information, Description: This command is used to verify that the system resources are functioning properly after the resolution.

Command: show system internal error, Expected Output: System Internal Error Information, Description: This command is used to verify that the internal system errors are resolved.

Command: show system power supply, Expected Output: System Power Supply Information, Description: This command is used to verify that the power supply configuration and status are functioning properly.
```

Roll-back Procedures

- Step: Restore the previous configuration, Command: copy /noappend running-config startup-config, Description: This step involves restoring the previous configuration. The copy command is used to copy the running configuration to the startup configuration.

- Step: Reload the system, Command: system reload no-wait, Description: This step involves reloading the system without waiting for user confirmation.